KNECK OFFICIAL

Chemistry — Paper 1

233/1 • ORIGINAL REVISION SET 01 • 2 hours • 80 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

Answer all questions. Show calculations and give units. Relative atomic masses: H=1, C=12, O=16, Na=23, Mg=24, S=32, Cl=35.5, Ca=40. Molar gas volume at room conditions: 24 dm³/mol.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

SHORT-ANSWER QUESTIONS

20 questions; 4 marks each.

1. A mixture contains iron filings, sodium chloride and sand. Describe how to obtain dry sodium chloride crystals and recover iron. [4 marks]

2. An element has atomic number 12 and mass number 24. State protons, neutrons, electron arrangement and usual ion charge. [4 marks]

3. Explain why molten sodium chloride conducts but solid sodium chloride does not. [4 marks]

4. Magnesium reacts with dilute hydrochloric acid. Write the balanced equation and describe a test for the gas. [4 marks]

5. Calculate mass of calcium carbonate needed to produce 4.8 dm³ CO2 at room conditions on complete acid reaction. [4 marks]

6. Distinguish a strong acid from a concentrated acid, and give one example of a strong acid. [4 marks]

7. Describe preparation of a dry sample of insoluble barium sulfate using suitable soluble salts. [4 marks]

8. Explain how increasing temperature increases the rate of a chemical reaction. [4 marks]

9. State two conditions for rusting and two ways to prevent it. [4 marks]

10. Balance Fe2O3 + CO → Fe + CO2, identify reducing agent and explain the choice. [4 marks]

11. Explain why diamond is hard and graphite conducts electricity. [4 marks]

12. Ethene is bubbled through bromine water. State observation, reaction type, product and one safety precaution. [4 marks]

13. Write the equation for complete combustion of ethanol; calculate moles CO2 from 0.5 mol ethanol. [4 marks]

14. State observations when aqueous sodium hydroxide is added dropwise then in excess to Cu^{2+} solution. Give ionic equation. [4 marks]

15. Explain why ammonia is collected by upward delivery and not usually over water. [4 marks]

16. Calculate pH of 0.001 mol/dm^3 HCl, assuming complete ionisation. State effect of tenfold dilution. [4 marks]

17. Electrolyse molten lead(II) bromide in a teacher demonstration. State products at each electrode and equations. [4 marks]

18. A hydrocarbon contains 85.7% carbon and 14.3% hydrogen by mass. Find empirical formula. [4 marks]

19. Describe how fractional distillation separates crude oil; name one fraction and its use. [4 marks]

20. Explain two environmental problems from burning sulfur-containing fossil fuels and one control measure. [4 marks]

END OF QUESTIONS

MARKING GUIDANCE

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Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

SHORT-ANSWER QUESTIONS

1. (4 marks) Use a magnet for iron (1); dissolve salt in water (1); filter off sand (1); evaporate some filtrate, cool to crystallise and dry (1).

2. (4 marks) 12 protons; 12 neutrons; 2.8.2; Mg^{2+} or +2. One each.

3. (4 marks) Ionic compound (1); ions fixed in lattice when solid (1); mobile ions when molten (1); mobile ions transport charge (1).

4. (4 marks) $Mg+2HCl \rightarrow MgCl_2+H_2$ (2); gas gives squeaky pop with a lighted splint, small sample away from reaction (2).

5. (4 marks) $n(CO_2)=4.8/24=0.2$ mol (1); 1:1 mole ratio (1); $Mr(CaCO_3)=100$ (1); mass=20 g (1).

6. (4 marks) Strong means extensively ionised in water (2); concentrated means many moles per unit volume, independent of strength (1); $HCl/HNO_3/H_2SO_4$ (1).

7. (4 marks) Mix aqueous $BaCl_2$ and Na_2SO_4 (1); $Ba^{2+}+SO_4^{2-} \rightarrow BaSO_4(s)$ (1); filter (1); wash residue with distilled water and dry (1).

8. (4 marks) Particles gain kinetic energy (1); collisions occur more frequently (1); larger fraction exceed activation energy (1); more successful collisions per unit time (1).

9. (4 marks) Water and oxygen (2). Paint/oil/galvanise/cathodic protection, any two (2).

10. (4 marks) $Fe_2O_3+3CO \rightarrow 2Fe+3CO_2$ (2); CO (1); removes oxygen from oxide/is oxidised to CO_2 (1).

11. (4 marks) Diamond has 4 strong covalent bonds per carbon (1) in a rigid three-dimensional network (1). Graphite has delocalised electrons (1) mobile along layers (1).

12. (4 marks) Orange/brown to colourless; addition; 1,2-dibromoethane; small-scale teacher-supervised fume-cupboard work/no skin contact. One each.

13. (4 marks) $C_2H_5OH+3O_2 \rightarrow 2CO_2+3H_2O$ (2); 1:2 ratio (1); 1.0 mol CO_2 (1).

14. (4 marks) Blue precipitate (1), insoluble in excess (1); $Cu^{2+}+2OH^- \rightarrow Cu(OH)_2(s)$ (2).

15. (4 marks) Less dense than air (1), displaces air downwards (1), very soluble in water (1), would dissolve instead of collect (1). Accept terminology with correctly described apparatus.

16. (4 marks) $[H^+]=0.001$ (1); $pH=-\log[H^+]$ (1); pH=3 (1); diluted pH=4 (1).

17. (4 marks) Cathode lead: $Pb^{2+}+2e^- \rightarrow Pb$ (2). Anode bromine: $2Br^- \rightarrow Br_2+2e^-$ (2). Toxic fumes require fume cupboard; not a home experiment.

18. (4 marks) C moles $85.7/12 \approx 7.14$ (1), H 14.3 (1), ratio 1:2 (1), CH_2 (1).

19. (4 marks) Vaporise mixture (1); condensation at different heights according to boiling ranges (1); valid fraction (1) and matched use (1), e.g. kerosene for jet fuel.

20. (4 marks) SO_2 forms acids in rain (1), damages vegetation/corrodes structures or acidifies lakes (1); particulates/CO_2 effect explained (1); desulfurisation/scrubbers/cleaner fuels (1).

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf